A. MAIN INPUT, COMMON BACKUP FUSE, TVS CLAMPS

B. BREAK-BEFORE-MAKE IDEAL-DIODE POWER PATHS

H01: RPS-400 CN3 +V to J101-1; CN2 -V to J101-2: 14 AWG.

Common-source MOSFET pairs block reverse and cross current.

J101 BOTTOM-PSU 24 V HARNESS

F101 15A

D101 SMCJ24A

R105 2.40R

V24_IN 1 2 GND

V24_IN V24_FUSED

GND V24_FUSED C101 1uF 50V X7R GND

V24_FUSED V24_FUSED 10uF 50V C106 GND

GATE_24 1 Q101 G S S S D 2 3 4 5 SRC_24

SRC_24 Q102 G S S S D 4 3 2 1 GATE_24

SRC_24 V24_SENSE

24 V operating window: UV fall 18.0 V, UV rise 20.0 V, OV rise 30.0 V

GATE_24 C110 47nF 50V SRC_24

BAT_SELECTED FROM LTC4418 PRESELECTOR

F201 15A

D201 SMCJ18A

R205 1.69R

BAT_SELECTED BAT_FUSED

GND BAT_FUSED C201

BAT_FUSED BAT_FUSED 1uF 50V X7R GND

BAT_FUSED BAT_FUSED 10uF 50V C205 GND

GATE_BAT 1 Q201 G S S S D 2 3 4 5 SRC_BAT

SRC_BAT Q202 G S S S D 4 3 2 1 GATE_BAT

SRC_BAT BAT_SENSE

Selected-backup outer window: UV fall 10.6 V, UV rise 11.6 V, OV rise 18.1 V

GATE_BAT C210 47nF 50V SRC_BAT

STARTING VALUE: ILIM = 25 mV / 1.50 mR = 16.7 A. Validate tolerance, SOA, and thermal margin before release.

C. LTC4421 PRIORITY CONTROLLER

R111 1.50mR KELVIN

V24_SENSE 3 4 RAW_OUT

R211 1.50mR KELVIN

BAT_SENSE 3 4 RAW_OUT

CPO C120 1uF 16V RAW_OUT

U101 LTC4421IUHE#PBF

RAW_OUT CPO RAW_OUT QUAL

32 33 34 35

V24_FUSED 3 UVF_24 4 UVR_24 5 OV_24 6 TMR_24 7 FAULT_DISABLE_24 8 CH_24V_N 9 VALID_24V_N 10 FAULT_DISABLE_24 11 BAT_FUSED 27 UVF_BAT 26 UVR_BAT 25 OV_BAT 24 TMR_BAT 23 FAULT_DISABLE_BAT 22 CH_BAT_N 21 VALID_BAT_N 20 FAULT_DISABLE_BAT 19

V1 UVF1 UVR1 OV1 TMR1 DISABLE1 CH1 VALID1 FAULT1 V2 UVF2 UVR2 OV2 TMR2 DISABLE2 CH2 VALID2 FAULT2

CPORDET CPO EXTVCC QUAL

SOURCE1 GATE1 SENSE1 OUT1 SOURCE2 GATE2 SENSE2 OUT2 CASOUT RETRY SHDN

1 SRC_24 2 GATE_24 36 V24_SENSE 35 RAW_OUT 29 SRC_BAT 28 GATE_BAT 30 BAT_SENSE 31 RAW_OUT 14 17 GND 18 SHDN_MAIN

INTVCC13 INTVCC GND EP GND GND CASIN INTVCC12

RETRY is grounded: a sustained overcurrent latches off until the selector is power-cycled.

INTVCC C121 1uF 16V GND

QUAL C122 1nF C0G GND

TMR_24 C123 100nF GND

TMR_BAT C223 100nF GND

C122 = 1 nF gives about 16 ms input qualification.

C123/C223 = 100 nF gives about 8.3 ms current-limit timer.

R311 1.00mR LOAD KELVIN

RAW_OUT 3 4 RAW_OUT_LOAD

R131 100k

INTVCC FAULT_DISABLE_24

R231 100k

INTVCC FAULT_DISABLE_BAT

H. USER SYSTEM POWER SWITCH

SW201: E-Switch RAB12C1121 maintained DPST OFF-ON panel rocker.

OFF opens both poles: 47k pull-downs disable LTC4421 and LTC4418 (fail-OFF).

J204 KEYED 4-WIRE SWITCH HARNESS

R541 47k

R542 47k

INTVCC 1 PRE_INTVCC 3 2 SHDN_MAIN 4 SHDN_PRE

SHDN_MAIN GND

SHDN_PRE GND

SW201 pole A: J204-1 INTVCC to J204-2 SHDN_MAIN.

SW201 pole B: J204-3 PRE_INTVCC to J204-4 SHDN_PRE.

Switch carries only controller-enable current; no 10 A load current crosses the panel.

D. RAW OUTPUT AND TRANSFER HOLD-UP

J301 RAW OUT TO LMS176

RAW_OUT_LOAD 1 2 RAW_OUT_LOAD GND 3 4 GND

RAW_OUT 220uF 50V GND C301

RAW_OUT 220uF 50V GND C302

RAW_OUT 220uF 50V GND C303

RAW_OUT 1uF 50V X7R GND C304

RAW_OUT 100nF 50V GND C305

RAW_OUT 6800uF 50V GND C306

RAW_OUT 6800uF 50V GND C307

RAW_OUT 6800uF 50V GND C308

RAW_OUT 6800uF 50V GND C309

C301-C303 are low-ESR Panasonic hybrid capacitors.

C306-C309: 4 x 6800 uF = 27.2 mF; total with C301-C303 is 27.86 mF nominal.

At ~20% capacitance the total provides about 25.4 ms ideal hold-up at 184.2 W from 24.0 V to 12.5 V.

x 35 mm cans require a tray-supported clamp; board outline and dock clearance remain A2 sample gates.

E. VALID-VOLTAGE WINDOWS (0.5 V COMPARATOR TAPS)

Physical order is VIN - R4 - UVF - R3 - UVR - R2 - OV - R1 - GND.

V24_FUSED UVF_24 R104 1.06M 0.1%

UVF_24 UVR_24 R103 3.01k 0.1%

UVR_24 OV_24 R102 9.10k 0.1%

OV_24 GND R101 18.2k 0.1%

BAT_FUSED UVF_BAT R204 965k 0.1%

UVF_BAT UVR_BAT R203 4.22k 0.1%

UVR_BAT OV_BAT R202 15.8k 0.1%

OV_BAT GND R201 28.0k 0.1%

24 V: OV=29.95 V, UV rising=19.97 V, UV falling=17.99 V

Battery: OV=18.09 V, UV rising=11.58 V, UV falling=10.57 V

F. BATTERY WARNING AND CARRIER STATUS

U301 TPS3842A015DRLRQ1

MON_3V3 6 VDD SENSE 5 BAT_MON 1 CTR GND 2 GND RESET 4 BAT_LOW_N CTS 3

J401 STATUS TO CARRIER

GND 1 CH_24V_N 2 CH_BAT_N 3 VALID_24V_N 4 VALID_BAT_N 5 BAT_LOW_N 6 VALID_DTAP_N 7 VALID_GOLD_N 8

BAT_FUSED BAT_MON R301 174k 0.1%

BAT_MON GND R302 10.0k 0.1%

MON_3V3 C310 100nF 10V GND

BAT_MON C311 1nF COG GND

Hardware backup-low policy: assert about 13.02 V; 5% hysteresis releases near 13.67 V.

J401 pins 1-8: GND, /CH24, /CHBAT, /VALID24, /VALIDBAT, /BATLOW, /VDTAP, /VGOLD

Open-drain outputs require 3.3 V pull-ups on the carrier.

I. PRIMARY / BACKUP / LOAD DIGITAL POWER TELEMETRY

INA228: 85 V, 20-bit shunt/bus monitor; common I2C with wired-OR active-low alert.

U601 0x40

MON_3V3 GND

MON24_VBUS 8 Vbus VS A1 1 A0 2 GND

MON24_INP 10 Vin+ SDA 4 PWR_MON_SDA

MON24_INN 9 Vin- SCL 5 PWR_MON_SCL

GND Alert 3 PWR_MON_ALERT_N

GND

U602 0x41

MON_3V3 GND MON_3V3

MONBAT_VBUS 8 Vbus VS A1 1 A0 2 GND

MONBAT_INP 10 Vin+ SDA 4 PWR_MON_SDA

MONBAT_INN 9 Vin- SCL 5 PWR_MON_SCL

GND Alert 3 PWR_MON_ALERT_N

GND

U603 0x44

MON_3V3 MON_3V3 GND

MONLOAD_VBUS 8 Vbus VS A1 1 A0 2 GND

MONLOAD_INP 10 Vin+ SDA 4 PWR_MON_SDA

MONLOAD_INN 9 Vin- SCL 5 PWR_MON_SCL

GND Alert 3 PWR_MON_ALERT_N

GND

V24_SENSE MON24_INP

RAW_OUT MON24_INN

MON24_INP MON24_VBUS

MON24_INN GND

V24_FUSED MON24_VBUS MON_3V3 GND

MON_3V3 1 GND 2 PWR_MON_SDA 3 PWR_MON_SCL 4 PWR_MON_ALERT_N 5 GND 6

J402 TELEMETRY: 1 3V3, 2 GND, 3 SDA, 4 SCL, 5 /ALERT, 6 GND

Source shunts: 1.5mR -> 27.307 A at +/-40.96 mV. Load shunt: 1mR -> 40.96 A.

BAT_SENSE MONBAT_INP

RAW_OUT MONBAT_INN

MONBAT_INP MONBAT_VBUS

MONBAT_INN GND

BAT_FUSED MONBAT_VBUS MON_3V3 GND

RAW_OUT MONLOAD_INP

RAW_OUT_LOAD MONLOAD_INN

MONLOAD_INP MONLOAD_VBUS

MONLOAD_INN GND

RAW_OUT_LOAD MONLOAD_VBUS MON_3V3 GND

PWR_FLAG MON_3V3

R311 is after the hold-up bank, so U603 includes capacitor-supplied load current during transfer.

G. D-TAP / GOLD MOUNT NON-PARALLELING PRESELECTOR

LTC4418 V1 gives valid D-Tap priority over V2 Gold Mount. Both paths are break-before-make.

External interface: LEMO EGG.0B.302.CLL female panel inlet: pin 1 return, pin 2 positive.

J202 FROM 2-PIN LEMO PANEL INLET 13.0-16.8 V

F202 15A

D202 SMBJ18CA

DTAP_IN 1 2 DTAP_IN DTAP_IN DTAP_FUSED GND 3 4 GND

GND DTAP_FUSED

DTAP_FUSED 10uF 50V X7R GND C202

GATE_DTAP Q501 G S S S D 4 3 2 1 SRC_DTAP

GATE_DTAP Q502 G S S S D 4 3 2 1 SRC_DTAP

DTAP_FUSED BAT_SELECTED

J203 FROM PANEL QRC-GOLD 8375-0094 HARNESS 10.0-16.8 V

D203 SMBJ18CA

GOLD_IN 1 2 GOLD_IN GOLD_IN GOLD_FUSED GND 3 4 GND

GND GOLD_FUSED

GOLD_FUSED 10uF 50V X7R GND C203

GATE_GOLD Q503 G S S S D 4 3 2 1 SRC_GOLD

GATE_GOLD Q504 G S S S D 4 3 2 1 SRC_GOLD

GOLD_FUSED BAT_SELECTED

SIR5607DP: -60 V P-MOS, 12 mOhm max at VGS=-4.5 V. Use large copper heat-spreading areas.

D-Tap and Gold Mount are alternate backup inputs only. Never parallel and never charge through this board.

REVERSE MISWIRE SAFE: fuse + bidirectional TVS + ceramic C precede LTC4418/back-to-back P-MOS blockers.

G1_CTRL GATE_DTAP

GATE_DTAP 47nF 50V SRC_DTAP 470nF 50V BAT_SELECTED BAT_SELECTED

G2_CTRL GATE_GOLD

GATE_GOLD 47nF 50V SRC_GOLD 470nF 50V BAT_SELECTED BAT_SELECTED

U201 LTC4418IUF#PBF

DTAP_FUSED 17 V1 UV1 2 OV1 3 VALID_DTAP_N 6 VALID1

GOLD_FUSED 16 V2 UV2 4 OV2 5 VALID_GOLD_N 7 VALID2

HYS 20

VS1 14 SRC_DTAP G1 13 G1_CTRL VS2 12 SRC_GOLD G2 11 G2_CTRL VOUT 15 BAT_SELECTED CAS 8 EN 18 PRE_INTVCC SHDN 19 SHDN_PRE TMR 1 TMR_PRE

INTVCC GND EP 21

PRE_INTVCC 9 100nF 16V GND

TMR_PRE 1nF C0G GND

BAT_SELECTED 100uF 35V GND

C532 = 1 nF: about 16 ms source validation

R511/R521 + C511/C521 limit higher-voltage-source inrush; verify SOA on prototype

D511/D521: anode at controller G pin, cathode at MOSFET gate for fast turn-off

DTAP_FUSED UV_DTAP R603 969k 0.1%

UV_DTAP OV_DTAP R602 26.1k 0.1%

OV_DTAP GND R601 5.64k 0.1%

D-Tap actual: UV rise 12.62 V, UV fall 12.25 V, OV rise 18.00 V

GOLD_FUSED UV_GOLD R506 909k 0.1%

UV_GOLD OV_GOLD R505 31.6k 0.1%

OV_GOLD GND R504 5.62k 0.1%

Gold actual: UV rise 11.69 V, UV fall 11.35 V, OV rise 17.74 V

DESIGN INTENT

Priority: internal PSU 24 V, then D-Tap, then Gold Mount. No source paralleling.

Both selectors are break-before-make: C301-C309 provide continuous energy to the carrier input.

SW201 OFF disables both selectors; no valid source can energize RAW_OUT. Charge batteries externally.

PWR_FLAG V24_FUSED PWR_FLAG BAT_FUSED PWR_FLAG RAW_OUT PWR_FLAG DTAP_FUSED PWR_FLAG GOLD_FUSED PWR_FLAG BAT_SELECTED PWR_FLAG RAW_OUT_LOAD PWR_FLAG GND

Battery inputs use LTC4418 plus back-to-back P-MOS reverse-polarity blocking
27.2 mF / 50 V hold-up bank is retained on this floor-mounted board
RAW_OUT_LOAD feeds the protected CM5-CARRIER regulator input
Fixed priority: internal PSU 24 V, then D-Tap, then Gold Mount dock

ProComm

Sheet: /
File: PowerSelector.kicad_sch

Title: Radxa CM5 ProComm 24 V / Battery Seamless Power Selector

Size: A2	Date: 2026-08-16	Rev: A1
KiCad E.D.A. 10.0.5-rc1		Id: 1/1